MEMORANDUM

**TO:** Prof. Racheal Adams
**FROM:** Kasin Ito-Sato, Nathan Liclican, Tristyn Kelly
**DATE:** September 24, 2023
**SUBJECT:** Water Filter Technical Memo

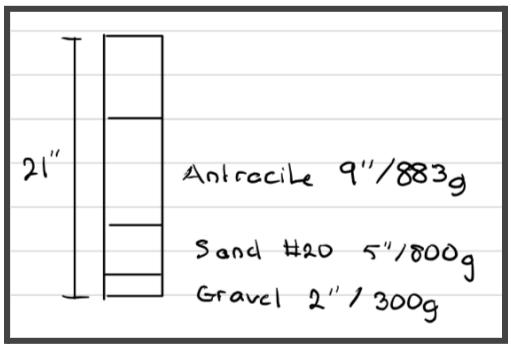
**Introduction:**

This unit centers on applying fundamental environmental engineering principles to create, assess, and enhance a basic water filtration system utilizing common granular filter materials like sand and anthracite. The group was tasked with analyzing experimental outcomes, refining the designs, and ultimately showcasing the findings. The goal of this project is to develop an understanding of the crucial role filtration plays in ensuring safe water quality.

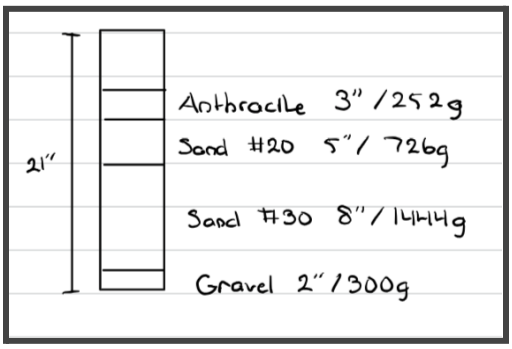
In this project, the group's filter design was assessed based on specific performance criteria. The researchers started with 1.2 liters of "dirty" water and were tasked with producing a minimum of 1 liter of "clean" water. The cleanliness of the water will be measured using turbidity, which quantifies how much suspended particles in water scatter light, making it cloudy. Turbidity is measured in nephelometric turbidity units (NTU) using a turbidimeter. The EPA's Safe Drinking Water Act mandates that nonconventional water must adhere to a turbidity standard of 5 NTU, while the conventional standard, preferred for safety, is 0.3 NTU. The researchers had the goal of getting the "dirty" water within the turbidity of 5 NTU.

To complete this project, the group created two different filter designs to get our water's NTU down to 5 NTU from the starting NTU. The researchers practiced using Kawamura's equation to find the effective grain size as the 10 percent size where 10% of the particles are smaller and can fit through the sieve.

**Filter Design & Methods:**




Filter 1:




Filter 2:

Filter 1: The original design was based on the idea that the filters used by large water filtering facilities are around a 1:2 ratio of sand to anthracite. The group came to a consensus to use soil #20 instead of soil #30 based on Kawamura's equation.

Filter 2: The second design was extremely sand-heavy because by observing the other groups' results we saw lower turbidity with the more sand used. The researchers also have much less anthracite because they felt that filtering out smaller particles is more difficult than the larger particles. This idea also made them choose to use soil #30 as well because it had a smaller density.

Challenges and Changes: The group's second filter yielded much stronger results compared to our original design. Adding more sand was successful in lowering the turbidity and it also increased the filtering time which could also contribute to the amount of particles removed. The first design yielded an average of 86 NTU and the second design yielded an average of 34.1 NTU.

**Performance Results & Discussion:**

| Filtrate Turbidity Test #1 (NTU)* | Filtrate Turbidity Test #2 (NTU) | Filtrate Turbidity Test #3 (NTU) | Time to Filter Test #1(min) | Time to Filter Test #2 (min) | Time to Filter Test #3 (min) |
|---|---|---|---|---|---|
| 31.9 | 83.1 | 88.9 | 2:12.44 | 1:02.27 | 1:01.32 |

| Inches of Anthracite | Inches of Sand (#20) | Inches of Sand (#30) | Inches of Gravel (Coarse Sand) | Total Media Height (inches) |
|---|---|---|---|---|
| 9" | 5" | 0" | 2" | 16" |

Filter 1:

| Filtrate Turbidity Test #1 (NTU)* | Filtrate Turbidity Test #2 (NTU) | Filtrate Turbidity Test #3 (NTU) | Time to Filter Test #1(min) | Time to Filter Test #2 (min) | Time to Filter Test #3 (min) |
|---|---|---|---|---|---|
| 30.1 | 34.4 | 33.8 | 6:15.78 | 5:04.18 | 5:42.14 |

| Inches of Anthracite | Inches of Sand (#20) | Inches of Sand (#30) | Inches of Gravel (Coarse Sand) | Total Media Height (inches) |
|---|---|---|---|---|
| 3 | 5 | 8 | 2 | 18 |

Filter 2:

The decrease in turbidity between the two tests further proved to the researchers that increasing the amount of sand used would improve the success of the filter. The first design yielded an average of 86 NTU and the second design yielded an average of 34.1 NTU. There was also an

extremely large increase in time between the two filters. Another factor that played a part in the difference in turbidity was the decrease in the amount of anthracite used. The group thought that the dust from the anthracite could be affecting the final turbidity results.

**Conclusion/Recommendation:**

The recommended filter design for a large-scale filter used in a facility would be different from our final design. The group's final design took into account that they are doing this filtering on a small scale and that they are only using the filter a few times. Larger filters are required to be backwashed and cleaned now and then and that's a factor that needs to be considered for the larger designs. A larger filter would most likely look much more similar to our initial design. The group's initial design accounted for the diameter of the material that correlated with the diameter of the anthracite used. It also used a lot of anthracite which larger filters use because it improves the length of time the filters can be used without being replaced. A large-scale filter would likely use a 1:2 ratio of sand to anthracite.

If the group were to create a third design we would most likely increase the amount of sand used further as well as use more of sand #30. The smaller diameter of sand would allow for a greater amount of particles to be removed and the larger amount would mean greater time to filter. If the filter was larger than 20" then we could get closer to the goal of 5 NTU. Another thing that they would do is potentially backwash it more than once or run clean water through it multiple times to clean the filter media. One problem that could come from an increase in the amount of sand used or a larger-sized filter would be that the filtering time could exceed the limit of 10 minutes.

**References:**


Kawamura's Equation

$$d_1 = d_2 \left(\frac{\rho_2 - \rho_w}{\rho_1 - \rho_w}\right)^{0.667}$$

D1 and D2 represent the effective size of the medium to use. P1 and P2 represent the density of the filter medium while Pw represents the density of water. This equation is used to establish the appropriate filter medium size. The group used this equation and came to the decision that sand #20 would be the best medium to use when it comes to filtering out particles.

Three Valleys Water Treatment

The Three Valleys Water Treatment Facility shows that the appropriate medium for a water filtration system consists mostly of anthracite. The researchers saw this and implemented it into the first design where we included a majority of anthracite. After seeing the results, the group realized that anthracite is only appropriate with long term use. Since the group's filter was only used short term, the effects were not as proficient as the group intended.

## Water Filter Test Results

| Filter #1 Design Results | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Filtrate Turbidity Test #1 (NTU)* | Filtrate Turbidity Test #2 (NTU) | Filtrate Turbidity Test #3 (NTU) | Time to Filter Test #1(min) | Time to Filter Test #2 (min) | Time to Filter Test #3 (min) | Inches of Anthracite | Inches of Sand (#20) | Inches of Sand (#30) | Inches of Gravel (Coarse Sand) | Total Media Height (inches) |
| Team #1 | 21.8 | 82.1 | 76.1 | 0:53.00 | 0:51.36 | 0:55.11 | 10" | 5" | 0" | 2" | 17" |
| Team #2 | 13.8 | 52.5 | 65.5 | 2:08 | 2:11 | 2:30 | 8" | 5.75" | 0" | 2" | 15.75" |
| Team #3 | 9.87 | 55.8 | 61.7 | 1:46 | 0:47 | 0:48.88 | 10" | 5" | 0" | 1" | 16" |
| Team #4 | 18.7 | 69.2 | 83.2 | 1:44.69 | 1:37.50 | 1:21.87 | 9.14" | 0" | 4.57" | 2.29" | 16" |
| Team #5 | 31.9 | 83.1 | 88.9 | 2:12.44 | 1:02.27 | 1:01.32 | 9" | 5" | 0" | 2" | 16" |
| Team #6 | 33.5 | 86.8 | 85.8 | 1:37.52 | 1:00.19 | 55.52 | 10' | 5in | 0" | 2" | 17" |
| Team #7 | 45.9 | 53.2 | 60.2 | 4.05 | 3.52 | 3.46 | 5.5" | 6" | 0" | 3" | 14.5" |
| Team #8 | 31.4 | 105 | 109 | 49.61 sec | 45.82 sec | 41.01 sec | 8" | 4.5" | | 3.5" | 16" |
| Team #9 | 56.5 | 76.5 | 72.2 | 2:10 minutes | 2:26 min | 2:14 min | 2" | 4" | 3" | 1" | 10" |
| Team #10 | 48.2 | 67.9 | 70.1 | 1;25.6 | 2;28 | 2:13 | 4" | 3.5" | 4" | 3" | 14.5" |

To look for improvements, the group confided in the test results for each individual group. After analyzing the data, the researchers realized that groups with more sand tended to have better results. The group used this information and implemented it into their design by using a majority of sand #30 and sand #20 as a filter medium. This greatly improved their results by greatly reducing the turbidity within the water.

**Appendices:**

